

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461218.6867 +/- 0.0087 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.046
Transit depth (Rp/Rs)^2	2.17 +/- 1.37 [%]
Orbital Inclination (inc)	86.34 +/- 2.92
Airmass coefficient 1 (a1)	0.00758 +/- 0.00061
Airmass coefficient 2 (a2)	3.0 +/- 1.67
Scatter in the residuals of the lightcurve fit is	8.09 %
Best Comparison Star	#2 - [151, 42]
Optimal Aperture	0.99
Optimal Annulus	4.75
Transit Duration (day)	3e-06 +/- 6.4e-05

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.147 ± 0.046 vs 0.1326 (deviation 0.31σ)
Duration fit vs published	nan d vs 0.0737 d (deviation nanσ)
Mid-time O-C	-3084.9 min
Depth SNR	3.2
Residual scatter	8.09 %

Light curve

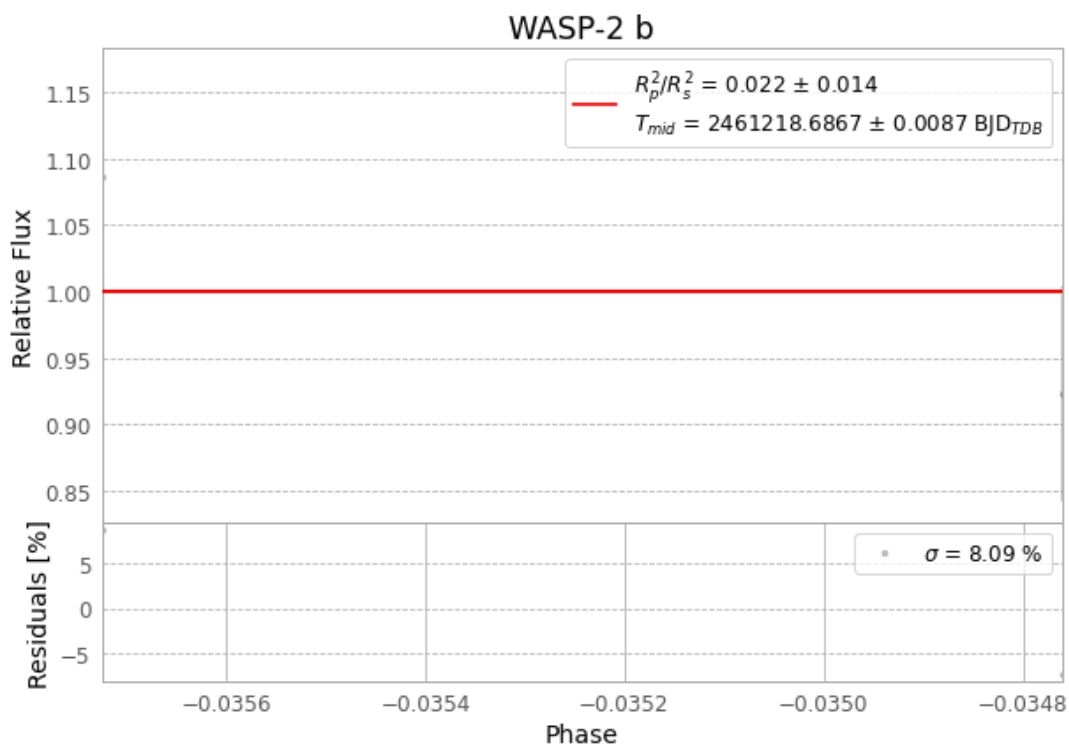


Figure 1 — FinalLightCurve_WASP-2 b_2026-06-29.png (SHA-256 0cde549bd785f033...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 182], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ed5a2b82cb8e2266...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

78 FITS frames (51 MB), WASP-2260629055207.FITS → WASP-2260629094215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-260629.log (afbfbba11a5c...), FinalLightCurve_WASP-2 b_2026-06-29.png (0cde549bd785...), FinalLightCurve_WASP-2 b_2026-06-29.csv (a3529ea5a011...)

Compute resources

Wall time 2104.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.